📈 DS REPORT: Bitcoin Market Sentiment & Trader Performance

Objective: To quantify the relationship between Hyperliquid trader performance and market sentiment (Fear & Greed Index) and deliver actionable trading strategies.

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1. Methodology Summary

The analysis involved merging two distinct datasets—daily market sentiment and granular, trade-level Hyperliquid data—and **aggregating performance metrics to the daily level** for direct comparison.

Key Analytical Steps:

1. **Data Alignment:** Standardized timestamps and merged both datasets on the daily date key.
2. **Feature Engineering:** Calculated crucial daily aggregates: **Average PnL**, **Profit Rate**, and **Daily Buy/Sell Ratio**.
3. **Statistical Validation:** Used the Kruskal-Wallis H-Test to formally test the dependency of PnL on sentiment.

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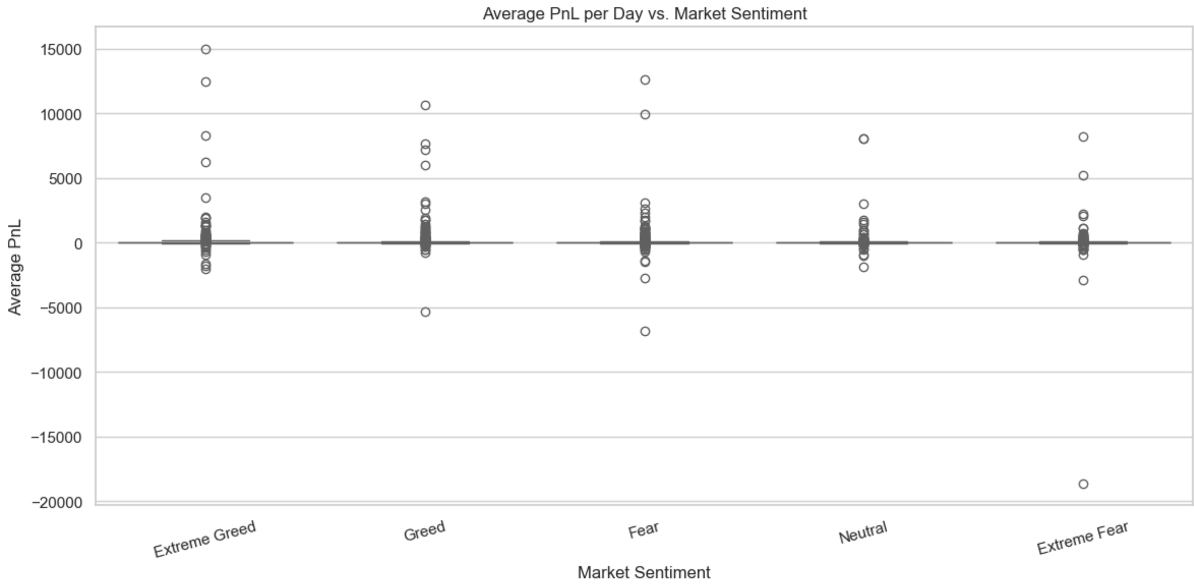
2. Key Findings & Statistical Validation

The analysis confirmed a **strong, statistically significant dependency** between market sentiment and collective trader performance.

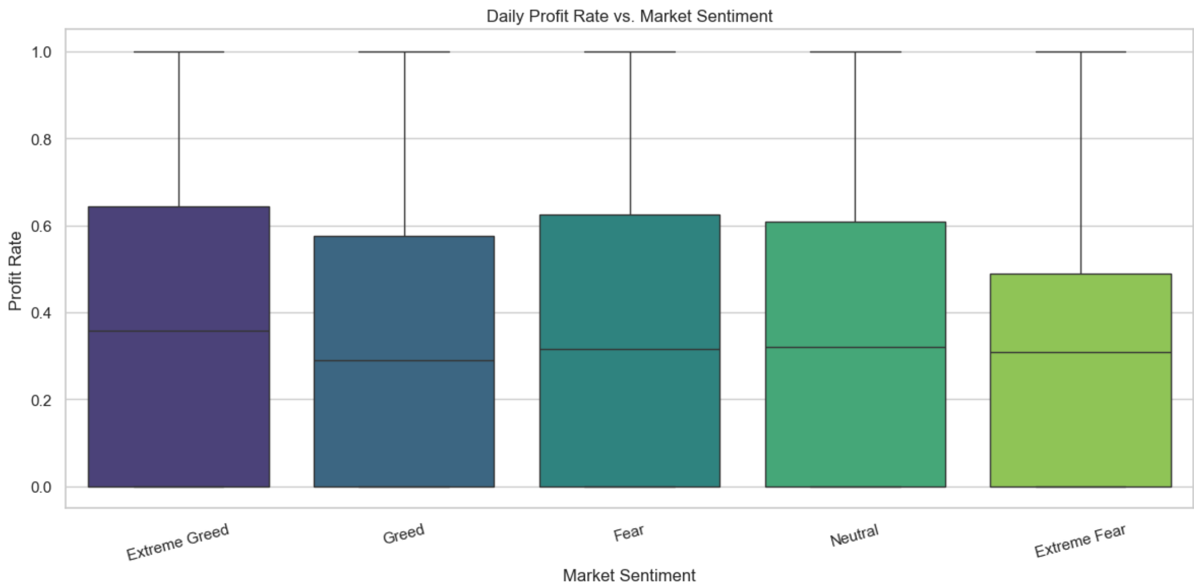
Finding 1: Performance is Statistically Dependent on Sentiment

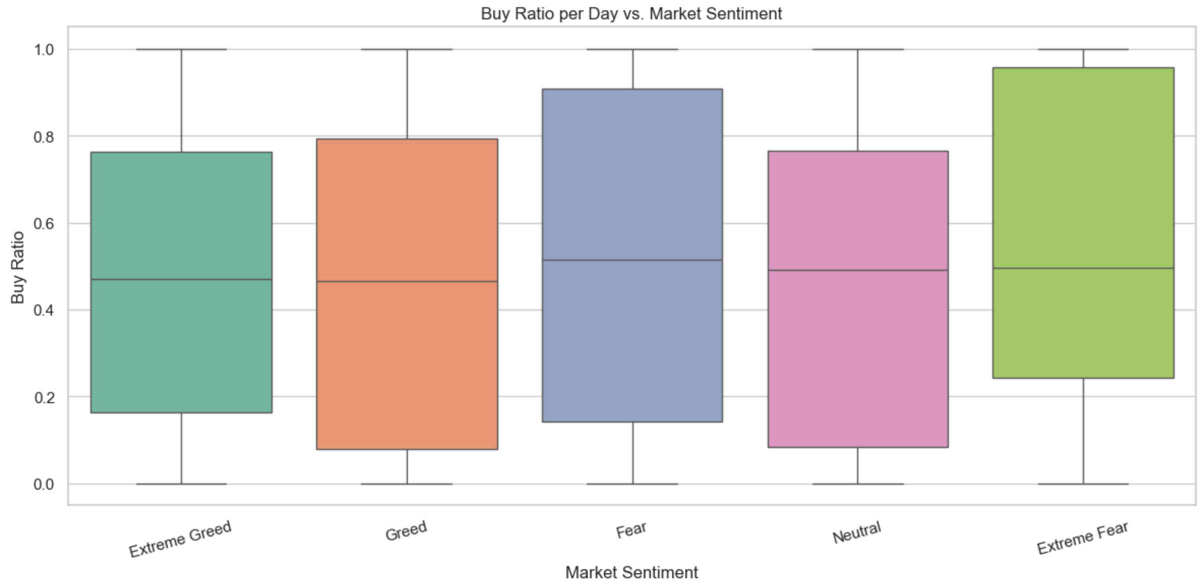
- **Observation:** Median trader performance (Avg PnL) is generally **highest during Greed** and **lowest during Extreme Fear** periods, highlighting a clear, sentiment-driven market bias.
- **Statistical Proof:** The Kruskal-Wallis H-Test yielded a **P-value of 0.00140**. Since $P<0.05$, we reject the null hypothesis and confirm that the difference in median PnL across the five sentiment classes is **highly significant**.



Finding 2: Behavioral Bias is Clear

- **Observation:** The **Profit Rate** is lowest during the **Fear** phase, suggesting that collective panic leads to poor execution. Traders exhibit **momentum-chasing behavior** with buy-side trades dominating during Greed.
- **Observation: Trade volume and size** increase significantly during Greed, indicating that traders increase their risk exposure when feeling optimistic.





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3. Actionable Strategy Recommendations

These insights can be used to build context-aware trading strategies, moving beyond simple price analysis.

🎯 Strategy 1: Inverse Risk Scaling

- **Logic:** The data shows traders take excessive risk during **Greed**.
- **Recommendation:** Implement an algorithm to **dynamically reduce leverage or position size** by 20–30% when sentiment hits **Greed** or **Extreme Greed**. This protects against sudden drawdowns caused by high collective leverage.

🎯 Strategy 2: Contrarian Trader Identification

- **Logic:** Some accounts consistently profit while the market panics (during Fear).
- **Recommendation:** Flag accounts that maintain a high **Profit Rate** while being net-short (low **Buy Ratio**) during **Extreme Fear**. These accounts represent potential "Smart Money" whose activity should be monitored for counter-trend signals.

🎯 Strategy 3: Sentiment-Driven Exit Logic

- **Logic:** A sharp shift in sentiment (e.g., from **Greed to Neutral**) can signal a temporary exhaustion of the trend.
- **Recommendation:** Use sentiment change as an **exit filter**. If a profitable long position is open and the market shifts from **Greed** to **Neutral**, trigger a partial profit-take or tighten the stop-loss to lock in gains.
